

Fractals

Infinite Details from Simple Rules

Speaker 1, Speaker 2, Speaker 3

CSE 200 Presentation

Department of Computer Science and Engineering
Bangladesh University of Engineering and Technology (BUET)

History

A Crisis in 19th-Century Mathematics

For centuries, calculus rested on an assumption that felt obvious:

The Assumption

Any continuous curve is **smooth almost everywhere** — zoom in far enough on any point, and the curve should straighten out into a well-defined tangent line.

A Crisis in 19th-Century Mathematics

For centuries, calculus rested on an assumption that felt obvious:

The Assumption

Any continuous curve is **smooth almost everywhere** — zoom in far enough on any point, and the curve should straighten out into a well-defined tangent line.

This wasn't just intuition — it was treated as close to self-evident. **Continuity** was assumed to imply **differentiability**, almost everywhere, almost always.

A Crisis in 19th-Century Mathematics

For centuries, calculus rested on an assumption that felt obvious:

The Assumption

Any continuous curve is **smooth almost everywhere** — zoom in far enough on any point, and the curve should straighten out into a well-defined tangent line.

This wasn't just intuition — it was treated as close to self-evident. **Continuity** was assumed to imply **differentiability**, almost everywhere, almost always.

In 1872, **Karl Weierstrass** broke this assumption — on purpose.

A Crisis in 19th-Century Mathematics

For centuries, calculus rested on an assumption that felt obvious:

The Assumption

Any continuous curve is **smooth almost everywhere** — zoom in far enough on any point, and the curve should straighten out into a well-defined tangent line.

This wasn't just intuition — it was treated as close to self-evident. **Continuity** was assumed to imply **differentiability**, almost everywhere, almost always.

In 1872, **Karl Weierstrass** broke this assumption — on purpose.

He constructed a function that is **continuous everywhere**, yet **differentiable nowhere**: infinitely jagged, at every single scale, with no tangent line anywhere at all.

Mathematical "Monsters"

Weierstrass's function was treated as a **pathological monster** — proof that pictures and intuition could not be trusted, and that analysis needed rigorous proof instead.

Mathematical "Monsters"

Weierstrass's function was treated as a **pathological monster** — proof that pictures and intuition could not be trusted, and that analysis needed rigorous proof instead.

This set off a chain reaction. Mathematicians began deliberately constructing stranger and stranger shapes, to probe exactly where classical geometry broke down:

Mathematical "Monsters"

Weierstrass's function was treated as a **pathological monster** — proof that pictures and intuition could not be trusted, and that analysis needed rigorous proof instead.

This set off a chain reaction. Mathematicians began deliberately constructing stranger and stranger shapes, to probe exactly where classical geometry broke down:

- **Georg Cantor** (1883) — the Cantor set: an infinite set of points with zero total length.

Mathematical "Monsters"

Weierstrass's function was treated as a **pathological monster** — proof that pictures and intuition could not be trusted, and that analysis needed rigorous proof instead.

This set off a chain reaction. Mathematicians began deliberately constructing stranger and stranger shapes, to probe exactly where classical geometry broke down:

- **Georg Cantor** (1883) — the Cantor set: an infinite set of points with zero total length.
- **Helge von Koch** (1904) — a geometric, drawable curve that is nowhere smooth: the **Koch curve**.

Mathematical "Monsters"

Weierstrass's function was treated as a **pathological monster** — proof that pictures and intuition could not be trusted, and that analysis needed rigorous proof instead.

This set off a chain reaction. Mathematicians began deliberately constructing stranger and stranger shapes, to probe exactly where classical geometry broke down:

- **Georg Cantor** (1883) — the Cantor set: an infinite set of points with zero total length.
- **Helge von Koch** (1904) — a geometric, drawable curve that is nowhere smooth: the **Koch curve**.
- **Wacław Sierpiński** — self-similar shapes like the **Sierpiński triangle**.

Mathematical "Monsters"

Weierstrass's function was treated as a **pathological monster** — proof that pictures and intuition could not be trusted, and that analysis needed rigorous proof instead.

This set off a chain reaction. Mathematicians began deliberately constructing stranger and stranger shapes, to probe exactly where classical geometry broke down:

- **Georg Cantor** (1883) — the Cantor set: an infinite set of points with zero total length.
- **Helge von Koch** (1904) — a geometric, drawable curve that is nowhere smooth: the **Koch curve**.
- **Wacław Sierpiński** — self-similar shapes like the **Sierpiński triangle**.
- **Felix Hausdorff** (1918) — gave these shapes a number: a **fractional dimension**, between the familiar integer dimensions.

Mathematical "Monsters"

Weierstrass's function was treated as a **pathological monster** — proof that pictures and intuition could not be trusted, and that analysis needed rigorous proof instead.

This set off a chain reaction. Mathematicians began deliberately constructing stranger and stranger shapes, to probe exactly where classical geometry broke down:

- **Georg Cantor** (1883) — the Cantor set: an infinite set of points with zero total length.
- **Helge von Koch** (1904) — a geometric, drawable curve that is nowhere smooth: the **Koch curve**.
- **Wacław Sierpiński** — self-similar shapes like the **Sierpiński triangle**.
- **Felix Hausdorff** (1918) — gave these shapes a number: a **fractional dimension**, between the familiar integer dimensions.
- **Gaston Julia & Pierre Fatou** (~1918) — studied iterated functions in the complex plane, without any way to actually see what they looked like.

Mathematical "Monsters"

Weierstrass's function was treated as a **pathological monster** — proof that pictures and intuition could not be trusted, and that analysis needed rigorous proof instead.

This set off a chain reaction. Mathematicians began deliberately constructing stranger and stranger shapes, to probe exactly where classical geometry broke down:

- **Georg Cantor** (1883) — the Cantor set: an infinite set of points with zero total length.
- **Helge von Koch** (1904) — a geometric, drawable curve that is nowhere smooth: the **Koch curve**.
- **Wacław Sierpiński** — self-similar shapes like the **Sierpiński triangle**.
- **Felix Hausdorff** (1918) — gave these shapes a number: a **fractional dimension**, between the familiar integer dimensions.
- **Gaston Julia & Pierre Fatou** (~1918) — studied iterated functions in the complex plane, without any way to actually see what they looked like.

Mandelbrot's Question

In the 1960s, working at IBM, **Benoit Mandelbrot** asked a deceptively simple question:

"How long is the coast of Britain?"

Mandelbrot's Question

In the 1960s, working at IBM, **Benoit Mandelbrot** asked a deceptively simple question:

"How long is the coast of Britain?"

The honest answer: **it depends on your ruler**. Measure with a 100 km ruler, get one length. Use a 1 km ruler, and the measured length grows — every bay and inlet you'd smoothed over now counts. The length never converges to a single number.

Mandelbrot's Question

In the 1960s, working at IBM, **Benoit Mandelbrot** asked a deceptively simple question:

"How long is the coast of Britain?"

The honest answer: **it depends on your ruler**. Measure with a 100 km ruler, get one length. Use a 1 km ruler, and the measured length grows — every bay and inlet you'd smoothed over now counts. The length never converges to a single number.

Mandelbrot's insight: this isn't a flaw in the coastline — it's a sign that classical length and area are **the wrong tools** for something like this. What a coastline actually needs is exactly the "useless" mathematics of Cantor, Koch, and Hausdorff.

Mandelbrot's Question

In the 1960s, working at IBM, **Benoit Mandelbrot** asked a deceptively simple question:

"How long is the coast of Britain?"

The honest answer: **it depends on your ruler**. Measure with a 100 km ruler, get one length. Use a 1 km ruler, and the measured length grows — every bay and inlet you'd smoothed over now counts. The length never converges to a single number.

Mandelbrot's insight: this isn't a flaw in the coastline — it's a sign that classical length and area are **the wrong tools** for something like this. What a coastline actually needs is exactly the "useless" mathematics of Cantor, Koch, and Hausdorff.

- He had a real problem no one had connected to these ideas.
- He had **computers** — something Cantor, Koch, and Julia never had — letting him see these shapes for the first time.

Definition

What are Fractals?

Definition

A **fractal** is a geometric shape containing detailed structure at arbitrarily small scales, usually having a **fractal dimension** strictly exceeding the **topological dimension**.

Key Characteristics:

- **Scale invariance:** Details at every level of magnification
- **Fractal dimension:** More than the *conventional* dimension
- **Infinite complexity:** Generated by simple, recursive rules
 - Like the infinite complexities of nature

What are Fractals?

Definition

A **fractal** is a geometric shape containing detailed structure at arbitrarily small scales, usually having a **fractal dimension** strictly exceeding the **topological dimension**.

Key Characteristics:

- **Scale invariance:** Details at every level of magnification
- **Fractal dimension:** More than the *conventional* dimension
- **Infinite complexity:** Generated by simple, recursive rules
 - Like the infinite complexities of nature

*"Clouds are not spheres, mountains are not cones,
coastlines are not circles..."*

— **Benoit Mandelbrot**

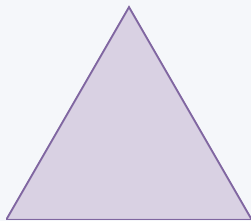
Some Fractals

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle



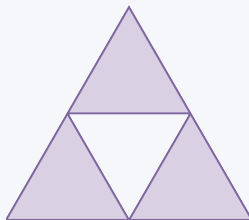
Iteration 0

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle
2. Remove the central triangle (triangle connecting midpoints of the sides)



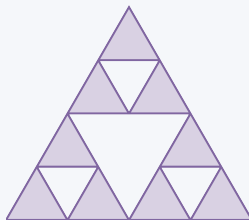
Iteration 1

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle
2. Remove the central triangle (triangle connecting midpoints of the sides)
3. Repeat for each triangle



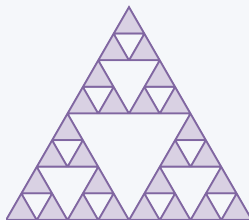
Iteration 2

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle
2. Remove the central triangle (triangle connecting midpoints of the sides)
3. Repeat for each triangle
4. After infinite iterations, we get the Sierpinski Triangle



Iteration 3

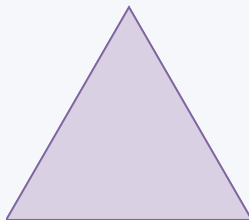
Area & Perimeter

The Sierpinski Triangle: Area

Now let's find the area of triangle(s):

Area Calculation

$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$



Iteration 0

The Sierpinski Triangle: Area

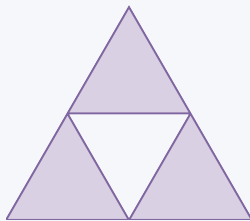
Now let's find the area of triangle(s):

Area Calculation

$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$

$$\begin{aligned} A_1 &= A_0 - \frac{1}{2} \cdot \sin(60^\circ) \cdot \left(\frac{a}{2}\right)^2 \\ &= A_0 - \frac{\sqrt{3}}{16} a^2 = A_0 - \frac{1}{4} A_0 \\ &= \frac{3}{4} A_0 \end{aligned}$$

We cut out one fourth of the area at each step.
So, naturally area is reduced by $\frac{3}{4}$ at each step.



Iteration 1

The Sierpinski Triangle: Area

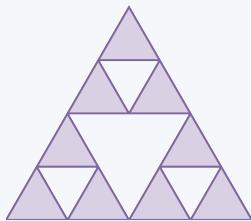
Now let's find the area of triangle(s):

Area Calculation

$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$

$$A_1 = \left(1 - \frac{1}{4}\right) A_0 = \frac{3}{4} A_0$$

$$A_2 = \left(1 - \frac{1}{4}\right) A_1 = \left(\frac{3}{4}\right)^2 A_0$$



Iteration 2

We cut out one fourth of the area at each step.
So, naturally area is reduced by $\frac{3}{4}$ at each step.

The Sierpinski Triangle: Area

Now let's find the area of triangle(s):

Area Calculation

$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$

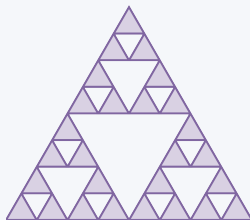
$$A_1 = \left(1 - \frac{1}{4}\right) A_0 = \frac{3}{4} A_0$$

$$A_2 = \left(1 - \frac{1}{4}\right) A_1 = \left(\frac{3}{4}\right)^2 A_0$$

...

$$A_n = \left(1 - \frac{1}{4}\right)^n A_0 = \left(\frac{3}{4}\right)^n A_0$$

$$\lim_{n \rightarrow \infty} A_n = 0$$



Iteration 3

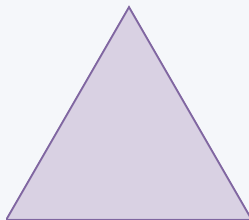
The area of the Sierpinski Triangle is zero even though it is a 2D shape.

The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

Perimeter Calculation

$$P_0 = 3a$$



Iteration 0

The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

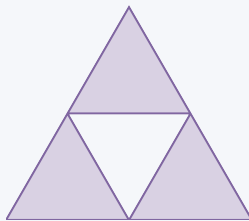
Perimeter Calculation

$$P_0 = 3a$$

$$P_1 = P_0 + 3 \cdot \frac{a}{2} = \frac{9}{2}a = \frac{3}{2}P_0$$

We add half of the previous perimeter at each step. This is because we add the same number of triangles as the last step, but each triangle is half the size of the previous one.

As a result, the perimeter grows by a factor of $\frac{3}{2}$ at each step.



Iteration 1

The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

Perimeter Calculation

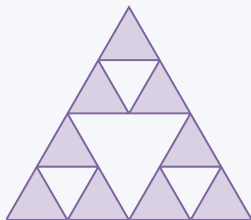
$$P_0 = 3a$$

$$P_1 = P_0 + 3 \cdot \frac{a}{2} = \frac{9}{2}a = \frac{3}{2}P_0$$

$$P_2 = P_1 + 3 \cdot 3 \cdot \frac{a}{4} = P_1 + \frac{1}{2}P_1 = \frac{3}{2}P_1$$

We add half of the previous perimeter at each step. This is because we add the same number of triangles as the last step, but each triangle is half the size of the previous one.

As a result, the perimeter grows by a factor of $\frac{3}{2}$ at each step.



Iteration 2

The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

Perimeter Calculation

$$P_0 = 3a$$

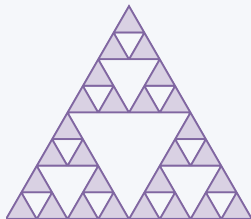
$$P_1 = P_0 + 3 \cdot \frac{a}{2} = \frac{9}{2}a = \frac{3}{2}P_0$$

$$P_2 = P_1 + 3 \cdot 3 \cdot \frac{a}{4} = P_1 + \frac{1}{2}P_1 = \frac{3}{2}P_1$$

...

$$P_n = \frac{3}{2}P_{n-1} = \left(\frac{3}{2}\right)^n P_0$$

$$\lim_{n \rightarrow \infty} P_n = \infty$$



Iteration 3

The Sierpinski Triangle has infinite perimeter but zero area.

Handoff to Speaker 2

We've now seen something strange, and precise:

- A **2D shape** (Sierpinski Triangle) with **zero area**.
- A shape built from a rule so simple a child could apply it by hand.
- No matter how far we zoom in, there's always more detail.

Handoff to Speaker 2

We've now seen something strange, and precise:

- A **2D shape** (Sierpinski Triangle) with **zero area**.
- A shape built from a rule so simple a child could apply it by hand.
- No matter how far we zoom in, there's always more detail.

If it isn't a line, and it isn't a surface — what **is** its true dimension?

Handoff to Speaker 2

We've now seen something strange, and precise:

- A **2D shape** (Sierpinski Triangle) with **zero area**.
- A shape built from a rule so simple a child could apply it by hand.
- No matter how far we zoom in, there's always more detail.

If it isn't a line, and it isn't a surface — what **is** its true dimension?

That number is exactly what Hausdorff was reaching for in 1918 — and it's where Speaker 2 picks up the story.

Fractal Dimension

Let's talk about a property of fractals. . .



Fractal Dimension

What does it mean for a shape to have a **non-integer** dimension?

Dimensions We Know

We are all familiar with integer dimensions:



Dimensions We Know

We are all familiar with integer dimensions:



Line



1D



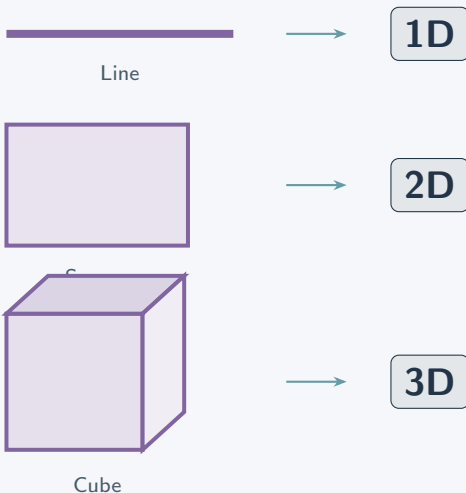
Square



2D

Dimensions We Know

We are all familiar with integer dimensions:



Dimensions are always whole numbers. . . right?

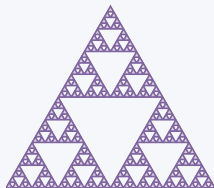
What About This One?

What dimension does this shape
have?

Any guesses?

- It lives in 2D space. . .
- But it has **zero area**
- And **infinite perimeter**

It's not quite 2D, but it's clearly more than
1D. . .



Sierpinski Triangle

What About This One?

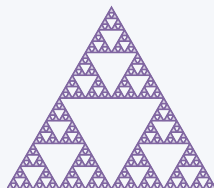
Sierpinski Dimension

The Sierpinski Triangle has a fractal dimension of:

$$D \approx 1.585$$

!!

A **non-integer** dimension — between a line (1D) and a filled surface (2D).



$$D = \log 3 / \log 2 \approx 1.585$$

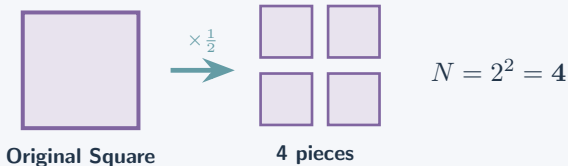
Self-Similarity & Scaling

Scaling a Line ($D = 1$): Scale by $r = \frac{1}{2} \implies 2$ pieces. Scale again by $\frac{1}{2} \implies 2 \times 2 = 4$ pieces.



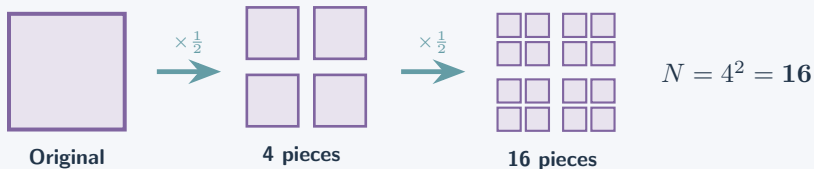
Self-Similarity & Scaling

Scaling a Square ($D = 2$) — Step 1: Scale by $r = \frac{1}{2} \implies$ **4 self-similar pieces** (2×2 grid).



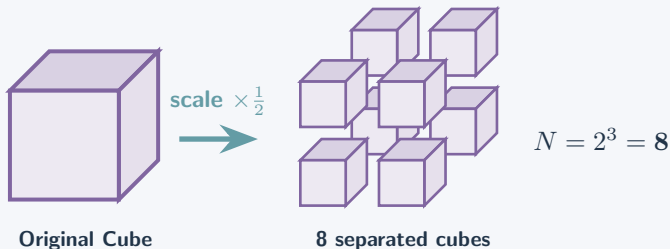
Self-Similarity & Scaling

Scaling a Square ($D = 2$) — Step 2: Scale each piece by $\frac{1}{2}$ again $\implies 4 \times 4 = \mathbf{16}$ self-similar pieces (4×4 grid)!



Self-Similarity & Scaling

Scaling a Cube ($D = 3$): Scale by $r = \frac{1}{2} \implies$
8 self-similar pieces ($2 \times 2 \times 2$ grid in 3D).



Self-Similarity & Scaling

Do You See a Pattern?

Each scaling step multiplies pieces by $(1/r)^D$!

Shape	Scale	1st Step	2nd Step	D
Line	$1/2$	$2 = 2^1$	$4 = 2^2$	1
Square	$1/2$	$4 = 2^2$	$16 = 2^4$	2
Cube	$1/2$	$8 = 2^3$	$(64 = 2^6)$	3

The Pattern

$$\text{Number of pieces} = \left(\frac{1}{r}\right)^D$$



$$D = ?$$

What if the shape is a **fractal**?

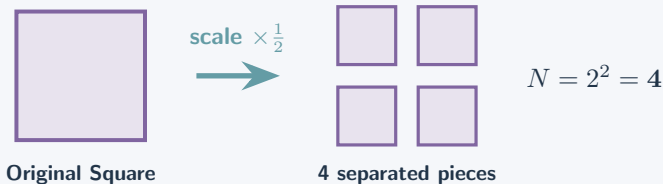
The Self-Similarity Rule

After scaling by factor r , the number of self-similar pieces is:

$$N = \left(\frac{1}{r} \right)^D$$

Square ($D = 2$) with scale factor $r = \frac{1}{2}$:

$$N = \left(\frac{1}{1/2} \right)^2 = 2^2 = 4 \text{ pieces}$$

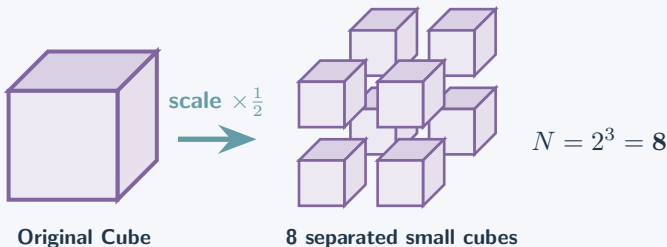


The Self-Similarity Rule

After scaling by factor r , the number of self-similar pieces is:

$$N = \left(\frac{1}{r} \right)^D$$

Cube ($D = 3$) with scale factor $r = \frac{1}{2}$: $N = \left(\frac{1}{1/2} \right)^3 = 2^3 = 8$ pieces



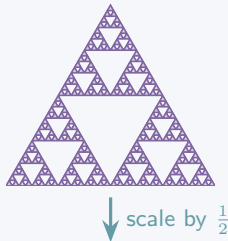
*Can we use this same rule to find the dimension of a **fractal**?*

Dimension of the Sierpinski Triangle

Let's apply the self-similarity rule to the **Sierpinski Triangle**.

Step 1

Scale the Sierpinski Triangle by $\frac{1}{2}$.

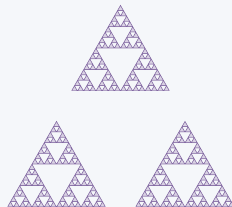


Dimension of the Sierpinski Triangle

Step 2 — Self-Similar Pieces

We get exactly **3** self-similar copies of the original triangle.

Each smaller triangle is a perfect replica, scaled by $r = \frac{1}{2}$.



3 copies

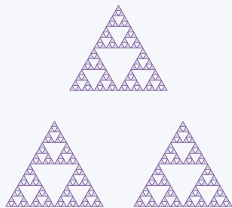
Dimension of the Sierpinski Triangle

Step 3 — The Equation

Using $N = \left(\frac{1}{r}\right)^D$:

$$3 = \left(\frac{1}{1/2}\right)^D$$

$$3 = 2^D$$



3 copies

Dimension of the Sierpinski Triangle

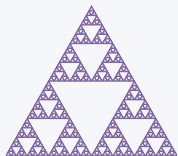
Step 4 — Solving for D

$$2^D = 3$$

$$D \log 2 = \log 3$$

$$D = \frac{\log 3}{\log 2}$$

$$D \approx \mathbf{1.585}$$



$$D \approx 1.585$$

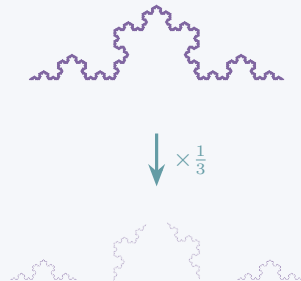
Not an integer! — strictly between 1D (line) and 2D (surface).

Similarly: The Koch Curve

The **Koch Curve** can be divided into **4** self-similar pieces, each scaled by $r = \frac{1}{3}$.

Setup

- Scale factor: $r = \frac{1}{3}$
- Self-similar pieces: $N = 4$



4 copies

Similarly: The Koch Curve

Koch Dimension

$$N = \left(\frac{1}{r}\right)^D$$

$$4 = 3^D$$

$$D = \frac{\log 4}{\log 3}$$

$$D \approx \mathbf{1.262}$$



$$D \approx 1.262$$

Also **non-integer** — between a line (1D) and a surface (2D), but closer to a line than the Sierpinski Triangle.

The General Theory: Box-Counting Dimension

The **Self-Similarity Dimension** works beautifully for self-similar fractals. But what about more complex shapes?

Box-Counting Dimension

The fractal dimension D is defined as:

$$D = \lim_{\varepsilon \rightarrow 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}$$

where $N(\varepsilon)$ is the number of boxes of side length ε needed to cover the fractal.

The General Theory: Box-Counting Dimension

The **Self-Similarity Dimension** works beautifully for self-similar fractals. But what about more complex shapes?

Box-Counting Dimension

The fractal dimension D is defined as:

$$D = \lim_{\varepsilon \rightarrow 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}$$

where $N(\varepsilon)$ is the number of boxes of side length ε needed to cover the fractal.

Intuition

- Cover the fractal with a grid of boxes of size ε
- Count how many boxes $N(\varepsilon)$ the fractal touches
- Shrink $\varepsilon \rightarrow 0$ and observe how $N(\varepsilon)$ grows
- The **rate of growth** tells us the dimension

Handoff to Speaker 3

We've now uncovered a remarkable property of fractals:

- They have **non-integer dimensions** — genuinely “between” the familiar 1D, 2D, 3D.
- The Sierpinski Triangle lives in dimension $D \approx 1.585$.
- The Koch Curve lives in dimension $D \approx 1.262$.
- The **box-counting dimension** generalises this to *any* shape.

Handoff to Speaker 3

We've now uncovered a remarkable property of fractals:

- They have **non-integer dimensions** — genuinely “between” the familiar 1D, 2D, 3D.
- The Sierpinski Triangle lives in dimension $D \approx 1.585$.
- The Koch Curve lives in dimension $D \approx 1.262$.
- The **box-counting dimension** generalises this to *any* shape.

We now know *what* fractals are and how to measure their complexity.

But **where do we see them?** And **how are they used** in computer science?

Handoff to Speaker 3

We've now uncovered a remarkable property of fractals:

- They have **non-integer dimensions** — genuinely “between” the familiar 1D, 2D, 3D.
- The Sierpinski Triangle lives in dimension $D \approx 1.585$.
- The Koch Curve lives in dimension $D \approx 1.262$.
- The **box-counting dimension** generalises this to *any* shape.

We now know *what* fractals are and how to measure their complexity.

But **where do we see them?** And **how are they used** in computer science?

That's exactly where Speaker 3 takes us — fractals in nature, art, and computation.

Fractals in the Real World

We know what a fractal is, and how to measure it. . .



Fractals in the Real World

Where does all of this actually **show up**?

Nature draws with one rule, repeated



Nature draws with one rule, repeated



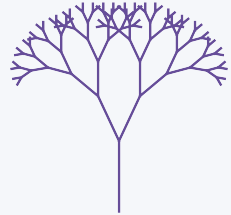
Nature draws with one rule, repeated



Nature draws with one rule, repeated



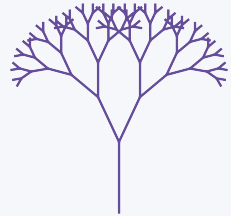
Nature draws with one rule, repeated



grown by a computer from
one rule: $F \rightarrow T[-F][+F]$



Nature draws with one rule, repeated



grown by a computer from
one rule: $F \rightarrow T[-F][+F]$



Broccoli, ice, leaves, cones: the **same shape at every zoom** — exactly what the Sierpinski rule did on paper.

A tree built from nothing but squares



Pythagorean tree, after A. E. Bosman (1942); symmetric at 45° , leaning at 55° .

A tree built from nothing but squares



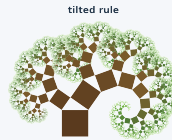
Pythagorean tree, after A. E. Bosman (1942); symmetric at 45° , leaning at 55° .

A tree built from nothing but squares



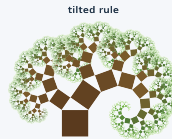
Pythagorean tree, after A. E. Bosman (1942); symmetric at 45° , leaning at 55° .

A tree built from nothing but squares



Pythagorean tree, after A. E. Bosman (1942); symmetric at 45° , leaning at 55° .

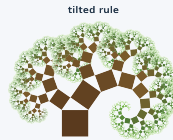
A tree built from nothing but squares



Stand a **right triangle** on each square's top edge and grow a new square on each leg — so $a^2 + b^2 = c^2$ sits at every single joint.

Pythagorean tree, after A. E. Bosman (1942); symmetric at 45° , leaning at 55° .

A tree built from nothing but squares



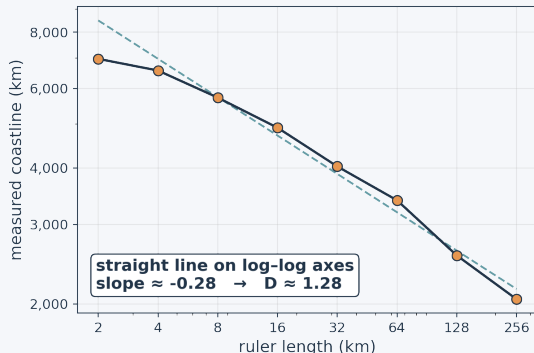
Stand a **right triangle** on each square's top edge and grow a new square on each leg — so $a^2 + b^2 = c^2$ sits at every single joint.

At 45° the two branches match exactly. **Tilt the triangle** and they stop matching — the tree leans, the way a real one grows toward the light.

Pythagorean tree, after A. E. Bosman (1942); symmetric at 45° , leaning at 55° .

Plot it on log-log axes: the slope is the dimension

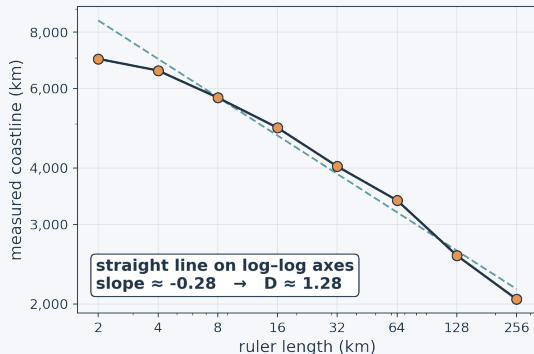
Walk a ruler of length ℓ along Britain's coast, add up the steps, then shrink ℓ and measure again.



Ruler-walk on the full-resolution GSHHG outline of Great Britain; rulers 4–256 km.

Plot it on log-log axes: the slope is the dimension

Walk a ruler of length ℓ along Britain's coast, add up the steps, then shrink ℓ and measure again.

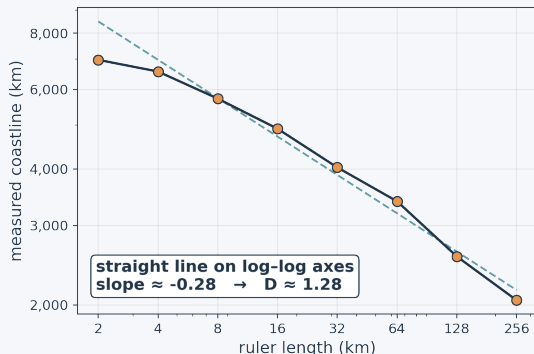


$$D \approx 1.3$$

Ruler-walk on the full-resolution GSHHG outline of Great Britain; rulers 4–256 km.

Plot it on log-log axes: the slope is the dimension

Walk a ruler of length ℓ along Britain's coast, add up the steps, then shrink ℓ and measure again.



$$D \approx 1.3$$

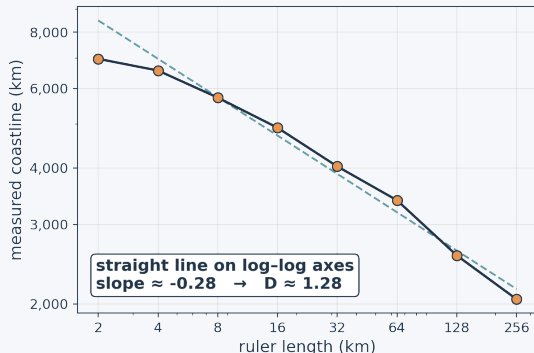
Mandelbrot (1967)

found 1.25

Ruler-walk on the full-resolution GSHHG outline of Great Britain; rulers 4–256 km.

Plot it on log-log axes: the slope is the dimension

Walk a ruler of length ℓ along Britain's coast, add up the steps, then shrink ℓ and measure again.



$$D \approx 1.3$$

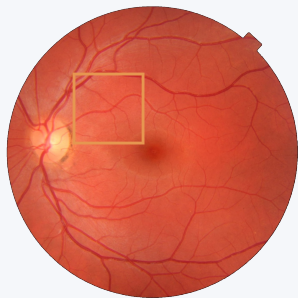
Mandelbrot (1967)

found 1.25

Speaker 2's box-counting idea, done with a ruler instead of boxes.

Ruler-walk on the full-resolution GSHHG outline of Great Britain; rulers 4–256 km.

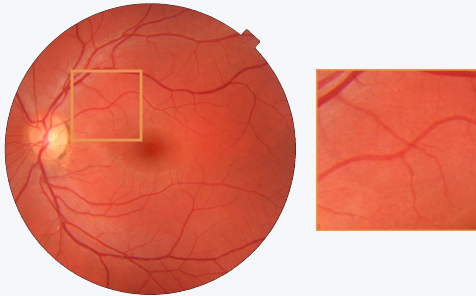
Your body branches to pack a huge surface into a small space



blood vessels in a human eye, and a zoom-in

Photo: M. Häggström, "Fundus photograph of normal left eye", 2014, CC0. Lung figures: Weibel (1963).

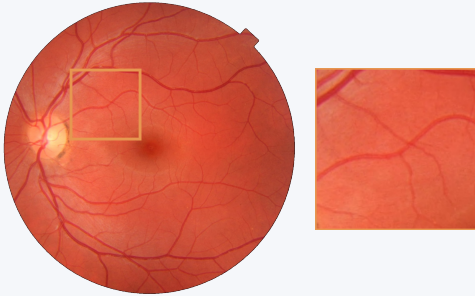
Your body branches to pack a huge surface into a small space



blood vessels in a human eye, and a zoom-in

Photo: M. Häggström, "Fundus photograph of normal left eye", 2014, CC0. Lung figures: Weibel (1963).

Your body branches to pack a huge surface into a small space

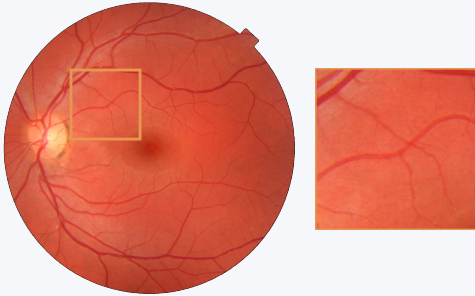


- Lungs: about **23 splits**
⇒ **$\sim 70 \text{ m}^2$** of surface
inside ~ 5 litres

blood vessels in a human eye, and a zoom-in

Photo: M. Häggström, "Fundus photograph of normal left eye", 2014, CC0. Lung figures: Weibel (1963).

Your body branches to pack a huge surface into a small space

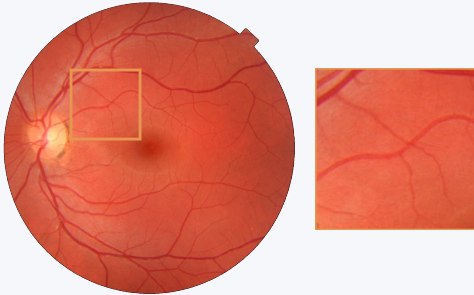


blood vessels in a human eye, and a zoom-in

- Lungs: about **23 splits**
 $\Rightarrow \sim 70 \text{ m}^2$ of surface
inside ~ 5 litres
- Vessels: every cell within
reach, using little space

Photo: M. Häggström, "Fundus photograph of normal left eye", 2014, CC0. Lung figures: Weibel (1963).

Your body branches to pack a huge surface into a small space

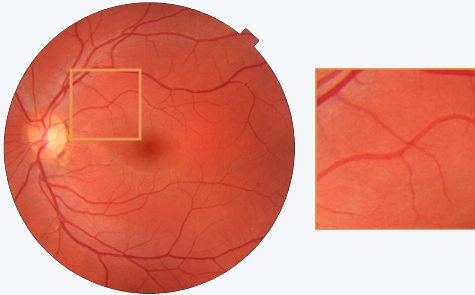


blood vessels in a human eye, and a zoom-in

- Lungs: about **23 splits**
 $\Rightarrow \sim 70 \text{ m}^2$ of surface
inside ~ 5 litres
- Vessels: every cell within
reach, using little space
- Doctors track the
dimension of eye vessels to
spot disease early

Photo: M. Häggström, "Fundus photograph of normal left eye", 2014, CC0. Lung figures: Weibel (1963).

Your body branches to pack a huge surface into a small space



blood vessels in a human eye, and a zoom-in

- Lungs: about **23 splits**
⇒ **~70 m²** of surface
inside ~5 litres
- Vessels: every cell within
reach, using little space
- Doctors track the
dimension of eye vessels to
spot disease early

Branching again and again is nature's cheapest way to get **area out of volume**.

Photo: M. Häggström, "Fundus photograph of normal left eye", 2014, CC0. Lung figures: Weibel (1963).

Fractal compression: a whole plant in 28 numbers

2,000 dots



Barnsley fern: an iterated function system (Barnsley, *Fractals Everywhere*, 1988), plotted by the chaos game.

Fractal compression: a whole plant in 28 numbers

25,000 dots



- Four rules, each a **shrink-and-tilt** of the whole picture, applied at random

Barnsley fern: an iterated function system (Barnsley, *Fractals Everywhere*, 1988), plotted by the chaos game.

Fractal compression: a whole plant in 28 numbers

200,000 dots



- Four rules, each a **shrink-and-tilt** of the whole picture, applied at random
- 4×6 coefficients + 4 probabilities = **28 numbers**

Barnsley fern: an iterated function system (Barnsley, *Fractals Everywhere*, 1988), plotted by the chaos game.

Fractal compression: a whole plant in 28 numbers

200,000 dots



- Four rules, each a **shrink-and-tilt** of the whole picture, applied at random
- 4×6 coefficients + 4 probabilities = **28 numbers**
- Those 28 numbers redraw the fern at **any resolution** — zoom in, there is always more leaf

Barnsley fern: an iterated function system (Barnsley, *Fractals Everywhere*, 1988), plotted by the chaos game.

Fractal compression: a whole plant in 28 numbers

200,000 dots



- Four rules, each a **shrink-and-tilt** of the whole picture, applied at random
- 4×6 coefficients + 4 probabilities = **28 numbers**
- Those 28 numbers redraw the fern at **any resolution** — zoom in, there is always more leaf

Store the **rule**, not the pixels — the idea behind **fractal image compression**.

Barnsley fern: an iterated function system (Barnsley, *Fractals Everywhere*, 1988), plotted by the chaos game.

Game worlds are grown from a rule, not stored

1 layer of bumps



Terrain generated here by summing 7 layers of noise, each half the size of the last (fractal Brownian motion).

Game worlds are grown from a rule, not stored

1 layer of bumps



3 layers



Terrain generated here by summing 7 layers of noise, each half the size of the last (fractal Brownian motion).

Game worlds are grown from a rule, not stored

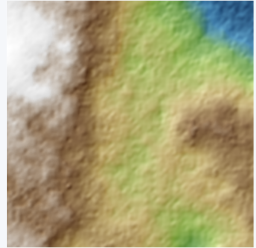
1 layer of bumps



3 layers



7 layers



Terrain generated here by summing 7 layers of noise, each half the size of the last (fractal Brownian motion).

Game worlds are grown from a rule, not stored

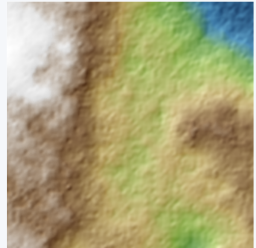
1 layer of bumps



3 layers



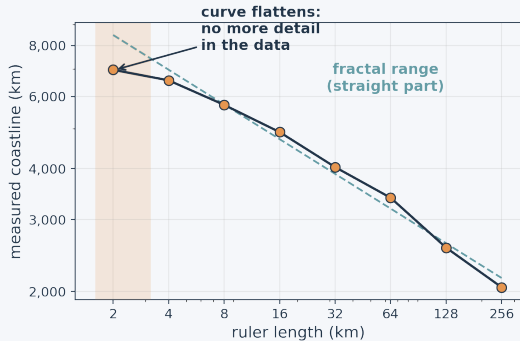
7 layers



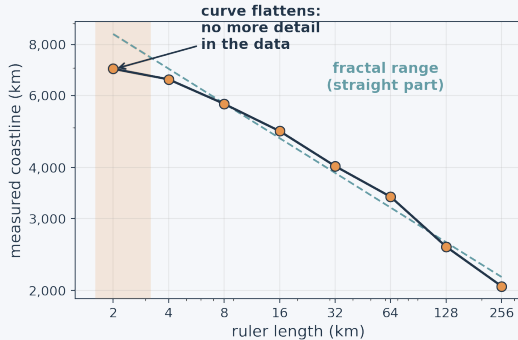
Add **smaller bumps at every scale** — the Koch idea — and a handful of numbers becomes an endless mountain range. Store the rule, not the map.

Terrain generated here by summing 7 layers of noise, each half the size of the last (fractal Brownian motion).

Real fractals stop at some scale

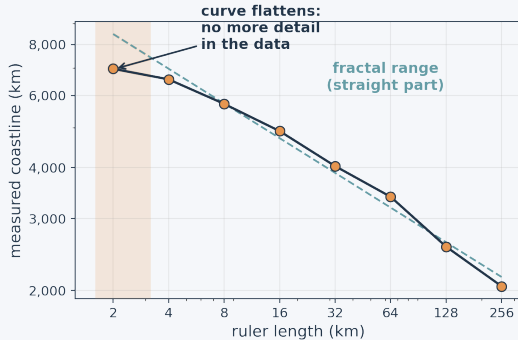


Real fractals stop at some scale



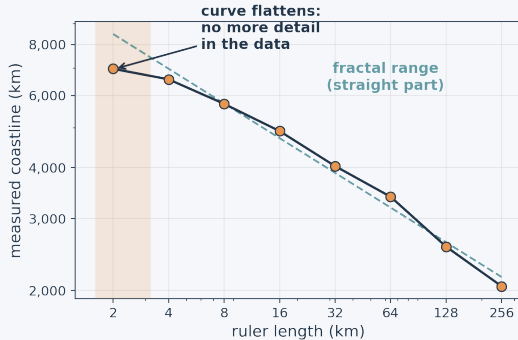
- Coast: bays look like bays, but **sand grains don't**

Real fractals stop at some scale



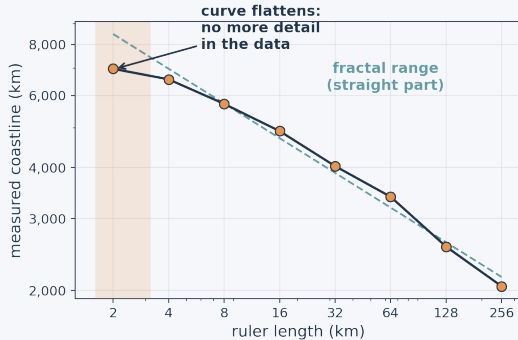
- Coast: bays look like bays, but **sand grains don't**
- Lungs: stop after ~23 splits

Real fractals stop at some scale



- Coast: bays look like bays, but **sand grains don't**
- Lungs: stop after ~23 splits
- Only the Sierpinski triangle goes on **forever**

Real fractals stop at some scale

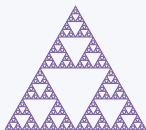


- Coast: bays look like bays, but **sand grains don't**
- Lungs: stop after ~23 splits
- Only the Sierpinski triangle goes on **forever**

A fractal is a **model** of nature — and every model has a range where it works.

One rule, repeated, explains what smooth geometry couldn't

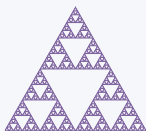
Part 1 — the surprise



zero area,
infinite edge

One rule, repeated, explains what smooth geometry couldn't

Part 1 — the surprise



zero area,
infinite edge

Part 2 — the number

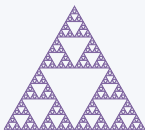
$$\frac{D = \log N}{\log(1/r)}$$

Sierpinski ≈ 1.585

Koch ≈ 1.262

One rule, repeated, explains what smooth geometry couldn't

Part 1 — the surprise



zero area,
infinite edge

Part 2 — the number

$$\frac{D = \log N}{\log(1/r)}$$

Sierpinski ≈ 1.585

Koch ≈ 1.262

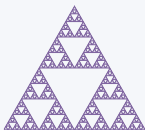
Part 3 — the world



trees, coasts, lungs,
ferns, game worlds

One rule, repeated, explains what smooth geometry couldn't

Part 1 — the surprise



zero area,
infinite edge

Part 2 — the number

$$\frac{D = \log N}{\log(1/r)}$$

Sierpinski ≈ 1.585

Koch ≈ 1.262

Part 3 — the world



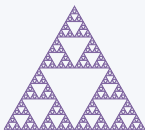
trees, coasts, lungs,
ferns, game worlds

*“Clouds are not spheres, mountains are not cones, coastlines
are not circles. . .” — Mandelbrot*

Roughness has a number, and simple rules make it.

One rule, repeated, explains what smooth geometry couldn't

Part 1 — the surprise



zero area,
infinite edge

Part 2 — the number

$$\frac{D = \log N}{\log(1/r)}$$

Sierpinski ≈ 1.585

Koch ≈ 1.262

Part 3 — the world



trees, coasts, lungs,
ferns, game worlds

*"Clouds are not spheres, mountains are not cones, coastlines
are not circles. . ." — Mandelbrot*

Roughness has a number, and simple rules make it.

Thank you — questions welcome.

References

- B. Mandelbrot, "How long is the coast of Britain? Statistical self-similarity and fractional dimension," *Science*, 1967.
- B. Mandelbrot, *The Fractal Geometry of Nature*, 1982.
- P. Wessel and W. H. F. Smith, "A global, self-consistent, hierarchical, high-resolution shoreline database," *J. Geophys. Res.*, 1996 (GSHHG data).
- E. R. Weibel, *Morphometry of the Human Lung*, Springer, 1963.
- M. Häggström, "Medical gallery of Mikael Häggström 2014," *WikiJournal of Medicine*, 2014 (retina photograph, CC0).
- M. F. Barnsley, *Fractals Everywhere*, Academic Press, 1988 (iterated function systems, the fern).
- A. E. Bosman, *Het wondere onderzoekingsveld van de vlakke meetkunde*, 1957 (Pythagorean tree, drawn 1942).
- A. Fournier, D. Fussell, L. Carpenter, "Computer rendering of stochastic models," *Comm. ACM*, 1982 (fractal terrain).
- 3Blue1Brown, "Fractals are typically not self-similar" (video).